SD-79 · Reconciliation & settlement engine

Fintech · staff (IC track) · full · standard · domain: fintech

2026-10-21 · attempt 1 on this question · round 6 overall · fictional sample candidate

Verdict

Outcome at the target level	Weighted score	Apparent level	Time outcome	Help needed
Hire (borderline)	**2.81 / 4**	staff-level depth on matching; senior-level framing	rushed the end — wrap took 4 of 9 min	(a) × 2 · (c) × 1

Headline. The matching design was genuinely strong — multi-pass, partitioned, with a clear story for corrected files — but the round treated reconciliation as *two lists that don't line up* rather than *expectations the ledger failed to meet*. That framing cost the fee/FX explanation at 00:41 and is the single change that moves this from a borderline hire to a clear one.

Scorecard

*Scored against the **staff** bar. Weights from `levels.md` §2 (staff): H = 1.5, N = 1.0, L = 0.5.*

#	Dimension	Weight	Score	Evidence
1	Requirements & scoping	N 1.0	3	00:06 asked who the users were before writing anything; cut real-time rails explicitly
2	Estimation	N 1.0	3	00:14 "15M × 15M is a cross product — partition by date, account, processor first" → drove the design
3	High-level design	N 1.0	3	clean raw → canonical → match → exceptions pipeline; walk-through at 00:22 without hesitation
4	Technical depth	**H 1.5**	3	many-to-one settlement matching and idempotent re-runs on corrected files, both two levels deep
5	Trade-off reasoning	**H 1.5**	3	00:31 rule-based vs ML matching: "rules first, because finance must explain every match"
6	Operations, reliability & cost	N 1.0	2	silent format change at 00:47 needed rung (a); no metric for unreconciled balance by age
7	Communication & collaboration	N 1.0	3	narrated continuously; defined "break" when asked, without condescension
8	Handling ambiguity & pushback	H 1.5	3	curveball (real-time rail) handled calmly; correctly said the core was batch-shaped
9	Leadership signal (technical)	N 1.0	2	ops-headcount question at 00:52 answered with "more automation" — no auto-match target, no sequencing

Weighted score = (3+3+3+4.5+4.5+2+3+4.5+2) ÷ 10.5 = 29.5 ÷ 10.5 = **2.81** → *hire* (2.75–3.24). No High-weight dimension at 1, so no cap applies. Dimension 6 is slightly contaminated by one interviewer leak (see self-audit) — read its 2 as "2, possibly generous", not as a reason to adjust the number.

Where the time went

Phase	Budget	Actual	Note
Requirements & scope	12	10	efficient; good clarifying questions
Estimation	9	6	short, but the one number mattered
High-level design	18	17	on budget
Deep dives	22	29	7 min on matching rules after the point was made
Wrap: failure, scale, cost, summary	9	4	four-sentence summary was three sentences and trailed off

Hint rungs used: (a) × 2 — format-change detection, ageing metric · (c) × 1 — “there’s a tension between what you expect and what arrives” at 00:41.

Retrospective

Your read — asked before any feedback

> “Matching felt solid. I think I lost time in the rules and didn’t land the ops story. I’d give myself a 3. The fee question surprised me — I hadn’t thought about the settlement being net.”

The interviewer’s read

Agree on matching and on time. The fee question was not a surprise to the design so much as a symptom: without ledger-generated expectations there is nowhere for fees, FX and chargebacks to live.

Calibration gap

Where you thought	Where the evidence put it	Why it matters
Overall ≈ 3	2.81 — borderline hire	close; slightly generous on ops
Lost time “in the rules”	yes, and it is what squeezed the wrap	the wrap is where dimensions 6 and 9 are earned
Fees were an edge case	fees, FX and chargebacks are the *normal* case for settlement	it is the crux of this question

Start · Stop · Continue

Start	Stop	Continue
Framing reconciliation as **expected vs actual** from the first minute	Elaborating a point after the interviewer has nodded it through	Asking who the users are before designing
Naming one ops metric per pipeline stage	Leaving the four-sentence summary to the last 60 seconds	Letting one estimate drive the partitioning decision

Feedback

Strengths

1. **Estimation that changed the design** — at 00:14, the cross-product observation immediately produced the partitioning scheme. Exactly the shape the bar asks for.

2. **Corrected files handled as first-class** — 00:35, “raw files are immutable, a resend is a new version, matches are recomputed for affected partitions.” Most candidates never get here.

Improvements

1. **The expectations model** — 00:41, a $3,400 shortfall had no explanation path. A strong answer has the ledger emit *expected settlements net of fees*, so the break is self-describing.
2. **The ops and people story** — 00:52, “more automation” for 25 ops staff. A Staff answer gives an auto-match target, the break taxonomy that drives upstream fixes, and the sequence.

Moments worth remembering

Time	Quote	Read
00:14	“15M by 15M is a cross product — I’m not doing that.”	altitude
00:31	“Rules first, because finance must be able to explain every match.”	strong trade-off
00:47	“We’d… notice when totals don’t match?”	needed rung (a)

The delta against a strong answer

What a strong answer contained that this one didn’t

1. **Ledger-generated expectations** including fees, FX and chargebacks — *mattered most*.
2. **Unreconciled balance by age** as the headline metric — mattered.
3. **Schema-change detection that fails loudly** — mattered; silent parse success is the classic incident.
4. Maker-checker on write-offs — checklist noise at this level; mentioned in passing is enough.

What you raised that the cue sheet doesn’t have

- Per-processor **confidence calibration** of fuzzy matches, sampled weekly by the ops team — a good idea worth a 4 on trade-offs had the rest been there.

The model design, in brief

Immutable raw files → canonical records → **expectations generated by the ledger** (amount, date, counterparty, net of known fees) → multi-pass matching (exact → rules → many-to-one) → breaks as a categorised, aged, owned workflow → net settlement under maker-checker. It hinges on one decision: **reconcile against expectations, not between two lists.**

Trend

Date	ID	Level	Weighted	Weakest dimension	Drill done?
2026-09-30	SD-44	staff	2.62	Ops (2)	yes
2026-10-07	SD-71	staff	2.95	Leadership (2)	partly
2026-10-14	SD-84	staff	3.10	Time (rushed)	yes
2026-10-21	SD-79	staff	2.81	Ops (2), Leadership (2)	—

Pattern: depth and trade-offs are reliably at the Staff bar. **Operations and the technical- leadership close are the standing weakness** — three of four rounds, and both are earned in the wrap, which keeps getting squeezed.

Next

The one drill. Run `wrap` mode three times this week on designs you already have, hard-stopping deep dives at their budget. Success = a crisp four-sentence summary and one named metric per stage.

Study the gap. Expected-vs-actual reconciliation; operational metrics for financial pipelines; how break taxonomies feed upstream fixes.

Next round. SD-81 *Payouts* at `staff` — it loads operations (ambiguous rail outcomes, returns) and rewards the expectations framing you're building.

Interviewer self-audit

One leak at 00:44: I said "net settlement" before the candidate did — dimension 6 slightly contaminated, flagged above rather than adjusted. Under-probed estimation (one probe only). Confidence in this scorecard: **medium-high**.